

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 4346	Course Title: Computer Hardware & Networking Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Students will be able to assemble and troubleshoot PCs/Laptops, install various drivers, operating system and application softwares, install and Configure Network Hardware Components. This course also helps the students to familiarize with network simulation tools and setting up of wired and wireless LANs.

Course Prerequisites:

Topic	Course name	Semester
Basic knowledge in Computer systems	Introduction to IT systems Lab	1
Combinational & sequential circuits	Logic System Design & Computer Organization	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate assembling and disassembling of desktop/laptop computers.	12	Applying
CO2	Installation and troubleshooting of Windows and Linux operating systems, drivers and various application softwares.	12	Applying
CO3	Installation and Configuration of Wired and Wireless Networks.	12	Applying
CO4	Open Ended Experiments**	6	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Assembling and disassembling desktop/ laptop computers.		
M1.01	Familiarize System Layout: Front panel indicators, switches, connectors and positions of SMPS, Motherboard, FDD, HDD, CD, DVD and add on cards.	1	Understanding
M1.02	Demonstrate various functional components in the motherboard, memory and processor.	2	Understanding
M1.03	Assembling and disassembling desktop PCs and laptops.	9	Applying
CO2	Install windows and Linux operating systems, drivers and various application softwares.		
M2.01	Demonstration of BIOS/CMOS Configuration (Date, Time, Enable/Disable Devices, Security, Dual BIOS, Booting sequence/Boot order	2	Applying
M2.02	Configure BIOS setup program and troubleshoot the typical problems using BIOS utility	2	Applying
M2.03	Partitioning & Formatting of Hard Drive - Primary, Extended, Logical partitions using Partition Tools.	2	Applying
M2.04	Installation of Operating System, drivers, Activation and Automatic Updating procedures.	3	Applying
M2.05	Installation and configuration of Application softwares, scanner, webcam, cell phone and biometric device with system.	3	Applying
	Lab Exam – I	1.5	

CO3	Installation and Configuration of Wired and Wireless Networks.		
M3.01	Demonstrate various hardware components used in Networking and its usage.	2	Applying
M3.02	Crimping of UTP Cable, Patch Panel Punching, Junction I/O Boxes, Optical Fibre Splicing.	3	Applying
M3.03	IP Addressing basics and Peer-to-Peer networking.	3	Applying
M3.04	Implement wired and wireless LANs.	4	Applying
CO4	Open Ended Projects.		
	Lab Exam – II	1.5	

**** - Suggested Open Ended Experiments**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

Sample Projects:

1. Analyse the type of network topology used in the lab and prepare a network diagram and technical specifications.
2. Implement a small wired/wireless network to share internet, printers, and files and prepare the network diagram and specifications.

Text / Reference

T/R	Book Title/Author
T1	Computer Hardware servicing & Networking - Dr.M.JEGANMOHAN,M.E.,MBA.,Ph.D.,(Management),Ph.D.,(Engg.),M.I.S.T.E.,
T2	A text book of Computer Hardware and Networking - Ms. Jyotika Deshmukh
R1	Computer Networking – A Top -Down Approach – James F Kurose& Keith W Ross
R2	Complete A+ Guide to IT Hardware and Software: AA CompTIA+Core 1(220-1001)&CompTIAL+Core 2(220-1002)textbook , 8th edition
R3	Comdex Hardware and Networking Course Kit Vikas Gupta DreamTech Press - 2011

Online Resources

Sl.No	Website Link
1	Learn Computer Hardware with Online Courses, Classes, & Lessons edX
2	https://onlinecourses.nptel.ac.in/noc23_cs35/preview
3	Computer Hardware Online Course for Free Learn to Earn (learnvern.com)